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# Evaluating termite community diversity through the S-W Index: A practical approach

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**Abstract-** The S-W Diversity Index, also known as Simpson's Diversity Index, is a metric used to quantify biodiversity within ecological studies. This paper presents a detailed methodology for calculating the S-W Diversity Index applied to termite populations. We illustrate the process using empirical data from a termite survey, highlighting the index's application and interpretation in assessing termite diversity. Our findings demonstrate how the S-W Diversity Index can provide insights into the ecological balance and species richness within termite communities. Termites play a crucial role in ecosystems as decomposers and soil engineers. Understanding their biodiversity is essential for ecological studies and conservation efforts. The S-W Diversity Index, derived from Simpson's Index, is a widely used tool for quantifying diversity, taking into account both species richness and abundance. The aim of this study is to Explain the calculation and significance of the S-W Diversity Index. Apply the index to termite population data to assess diversity. Discuss the implications of the diversity index in ecological research and management. Ten locations with 90 quadrates each were chosen as study points or sampling stations for the current experiment, and these sites were thoroughly examined to collect termite samples. Termite species were identified, and the abundance of each species was recorded. The S-W diversity values of the six termite species in this zone-specific habitat vary from 0.7114 to 0.6825 (January to December), with a maximum value in July of 2022 of 0.7434, and from 0.7473 to 0.7257 (January to December) with a maximum value in February of 2023 of 0.8021, based on data gathered from the various sampling sites. This value indicates a moderate level of diversity within the termite community. A diversity index suggests a reasonably diverse community, though not highly diverse. The balance between species richness and abundance influences the index, providing insights into ecological interactions and the stability of the termite population. Understanding termite diversity helps in assessing ecosystem health and planning conservation strategies. A higher diversity index may indicate a more stable and resilient ecosystem, while a lower index could signal environmental stress or imbalance. The S-W Diversity Index is a valuable tool for measuring biodiversity. In our study of termite populations, the index provided meaningful insights into community diversity. Future research should explore broader datasets and additional ecological factors to enhance our understanding of termite biodiversity and its implications for ecosystem management.

**Key words: S-W Diversity Index, termite, biodiversity, ecosystem health, species richness, abundance**

## INTRODUCTION

Biodiversity plays a fundamental role in maintaining ecological balance, and measuring species diversity is crucial for understanding ecosystem health.[1] Termites, as key decomposers and soil engineers, contribute significantly to nutrient cycling and habitat stability.[2] Their diversity and population dynamics serve as essential indicators of ecosystem function and environmental changes.

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The Shannon-Weaver (S-W) Diversity Index, derived from Simpson's Diversity Index,[3] is a widely used metric for quantifying biodiversity. It considers both species richness (the number of species present) and species abundance (the relative population size of each species), providing a comprehensive measure of community diversity. By applying the S-W Index to termite populations, researchers can assess variations in species distribution and ecological interactions across different habitats.

This study aims to evaluate termite community diversity using the S-W Diversity Index, providing a practical approach to its calculation and interpretation. Data were collected from ten study locations, each containing 90 quadrates, and termite species were identified and recorded. The diversity index values, which fluctuated between 0.7114 and 0.8021 over different months, indicate a moderate level of termite diversity. Understanding these variations helps in assessing the stability, resilience, and health of the ecosystem.[4]

By analyzing termite diversity, this research contributes to a broader understanding of biodiversity conservation and ecological management. The findings highlight the importance of monitoring termite populations to detect environmental changes, which can aid in sustainable ecosystem management strategies.

## MATERIALS & METHODS

### Study Area and Sampling Design

The study was conducted at ten selected locations within diverse habitats to evaluate termite community diversity. At each location, 90 quadrates (sampling units) were established for systematic sampling.[5] The quadrates were distributed strategically to ensure comprehensive coverage of the habitat and to capture variations in termite distribution.

### Sample Collection

Termite samples were collected manually and using a standardized soil sampling method. The collection involved:

- **Excavating soil and inspecting decaying wood, leaf litter, and logs** for termite presence.
- **Using forceps and brushes** to carefully transfer termites into labeled collection vials.
- **Recording environmental parameters**, such as temperature and soil moisture, to assess their influence on termite diversity.

### Species Identification and Abundance Recording

Collected termite specimens were preserved in 80% ethanol for further identification. Species were identified using standard taxonomic keys and reference books.[6-9] The abundance of each species was recorded to calculate species richness and relative frequency.

### Calculation of the S-W Diversity Index

The Shannon-Weaver (S-W) Diversity Index was used to assess termite biodiversity. It was calculated using the formula:

$$H' = -\sum (pi \ln pi)$$

where:

H' = Shannon-Weaver Diversity Index

$pi$ = Proportion of individuals of species $i$ relative to the total number of individuals

$\ln pi$ = Natural logarithm of $pi$

The index was computed for each sampling site and across different months to analyze temporal variations in diversity.

### Data Analysis

The diversity index values were statistically analyzed to determine seasonal trends in termite diversity. Variations in diversity were compared across different months and locations. The highest and lowest S-W Diversity Index values were recorded, along with their ecological significance.

## RESULTS & DISCUSSION

### Termite Species Diversity and Abundance

A total of six termite species were identified across the ten sampling locations. Species richness and abundance varied throughout the study period, reflecting seasonal and environmental influences on termite distribution. The recorded S-W Diversity Index values ranged from 0.6825 to 0.8021, indicating a moderate level of termite community diversity.

### Seasonal Variation in Termite Diversity

- In 2022, the S-W Diversity Index ranged between 0.7114 (lowest) and 0.7434 (highest), with peak diversity observed in July 2022.
- In 2023, the diversity index showed an increase, ranging between 0.7257 and 0.8021, with the highest value recorded in February 2023.
- The lowest diversity values were recorded during winter months, while the highest diversity was observed in monsoon and early summer.

**Biomass and Net Production Trends**

- The maximum biomass recorded was 4.858 g/m³, while the minimum was 2.631 g/m³.
- The biomass per haul varied from 1.3 g/haul (October 2022) to 2.4 g/haul (June–July 2022).
- The highest monthly net production was 0.608 g/m³/month (February 2023), and the highest daily net production was 0.018 g/m³/day (March 2023).
- Negative growth trends were observed during post-monsoon months, likely due to habitat changes and resource availability.

**Ecological Implications**

- The moderate diversity index suggests a stable termite community with a balanced species distribution.
- Seasonal fluctuations indicate that temperature, humidity, and food availability play significant roles in shaping termite diversity.
- A higher diversity index in monsoon and summer suggests favorable environmental conditions, while lower diversity in winter highlights the impact of cold temperatures on termite activity.

**Summary of Findings**

- The study confirms a direct correlation between temperature and termite productivity, with higher temperatures supporting greater species richness and abundance.
- The S-W Diversity Index proved effective in quantifying termite biodiversity, offering valuable insights for ecosystem monitoring and conservation planning.
- The results align with previous studies, indicating that termite diversity serves as a reliable indicator of ecosystem health and stability.

The findings of this study provide valuable insights into the diversity and ecological significance of termite communities within the study area. The S-W Diversity Index values, ranging from 0.6825 to 0.8021, indicate a moderate level of biodiversity, which suggests a relatively stable termite population. The observed seasonal variations in diversity emphasize the influence of environmental factors such as temperature, humidity, and food availability on termite distribution and productivity.

**Factors Influencing Termite Diversity**

**1. Seasonal Variation**

- The study revealed higher diversity indices during monsoon and early summer, with a peak in July 2022 (0.7434) and February 2023 (0.8021). This trend aligns with the increased availability of moisture and decomposing organic matter, which favors termite foraging and reproduction.
- In contrast, lower diversity indices were recorded during winter months, reflecting reduced termite activity due to lower temperatures. This finding is consistent with previous research indicating that termites are temperature-sensitive and exhibit reduced metabolic activity in cold conditions.

**2. Temperature and Habitat Conditions**

- A positive correlation between temperature and termite productivity was observed, supporting earlier studies that suggest temperature significantly affects termite abundance and species richness.
- The study area experiences habitat fluctuations, particularly drying during summer, which may contribute to increased termite activity in specific zones, leading to higher diversity values.

**3. Food Availability and Substrate Quality**

- Termites are primary decomposers, relying on decaying plant material and soil organic matter. The availability of wood, leaf litter, and moisture-rich habitats directly influences their abundance.
- The study suggests that habitat heterogeneity and the quality of organic matter in different sampling sites contribute to variations in termite diversity.

**4. Ecological Significance of Termite Diversity**

- A moderate to high diversity index suggests that termite communities are relatively stable and resilient in the study area.
- Termites play a crucial role in ecosystem processes, including nutrient cycling, soil aeration, and organic matter decomposition. Their population dynamics can serve as indicators of ecosystem health.
- A higher diversity index may indicate a well-balanced ecosystem, whereas a lower index could signal environmental stress, habitat degradation, or reduced resource availability.

**Comparison with Previous Studies**

The findings of this study align with previous research on termite biodiversity, supporting the role of temperature, substrate quality, and moisture availability as key drivers of termite diversity. Studies conducted by Benke & Benke (1975)[11] and Winberg (1971)[12] have also

demonstrated similar relationships between seasonal changes and insect productivity, further validating the observed trends.

**Implications for Conservation and Management**

- Monitoring termite diversity can serve as an early indicator of environmental changes and help assess the impact of climate variability, deforestation, and habitat fragmentation on ecosystem stability.
- Sustainable land-use practices should focus on preserving natural termite habitats to maintain their ecological functions.
- Future research should expand on long-term biodiversity monitoring, considering additional environmental variables such as soil composition, microclimatic conditions, and interspecific interactions to gain deeper insights into termite community dynamics.

## CONCLUSION

The study successfully applied the S-W Diversity Index to evaluate termite diversity, demonstrating its effectiveness in quantifying species richness and abundance. The findings highlight the importance of environmental factors in shaping termite populations and reinforce the need for ongoing monitoring to ensure ecosystem sustainability. The study provides a foundation for further ecological assessments and emphasizes the role of termites as key bioindicators in biodiversity conservation.

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